Intravenous Immunoglobulin (IVIG) and Aspirin in Kawasaki Syndrome

No definitive prospective study has demonstrated that high-dose aspirin (ASA) therapy prevents coronary artery abnormalities. In contrast, several randomized multicenter studies have shown that high-dose intravenous immunoglobulin (IVIG) (greater than 1 g/kg) in combination with ASA is safe and effective in reducing the prevalence of coronary artery abnormalities. IVIG reduces both the overall incidence of coronary artery abnormalities and the formation of giant aneurysms—the most serious coronary complication of Kawasaki syndrome (KS). Additionally, abnormalities in left ventricular systolic function and contractility improve more rapidly in children treated during the acute phase with high-dose IVIG plus ASA compared to those treated with ASA alone.

The efficacy of IVIG in preventing coronary artery abnormalities is dose-dependent, although the optimal dose has not been definitively established. Administration of a single dose of IVIG at 2 g/kg over 10 to 12 hours has been shown to be superior to the earlier regimen of 400 mg/kg per day for 4 days, with more rapid defervescence and reduced coronary artery abnormalities at 2-week follow-up (though no statistically significant difference is observed at 7 weeks). The current standard of care is administration of 2 g/kg IVIG plus ASA before day 10 of fever, which reduces the risk of coronary artery disease from approximately 20% to less than 5%. The precise mechanism of action of IVIG remains unknown.

Approximately 85% of children treated within 10 days of illness onset experience prompt defervescence and resolution of inflammatory signs following IVIG and ASA therapy. However, optimal management is not well defined for children who have persistent or recrudescent fever and ongoing signs of inflammation 24 to 72 hours or longer after initial treatment.

▲ FIGURE 168-5 Approach to the patient with suspected Kawasaki syndrome (KS). ASA = aspirin; dx = diagnosis; IVIG = intravenous immunoglobulin.

For patients in whom the diagnosis of KS remains likely and who remain symptomatic after one dose of IVIG, a second dose of IVIG (2 g/kg) is generally administered, as persistent fever correlates with elevated proinflammatory cytokines and an increased risk of coronary artery ectasia or aneurysm formation. In some cases, a third dose of IVIG may be considered, although complications have been reported with high cumulative doses of γ globulin.

If fever persists after two or three doses of IVIG, intravenous methylprednisolone (pulse therapy with 30 mg/kg per day for 3 days) may be considered. However, the use of systemic corticosteroids in KS remains controversial. Several Japanese studies have reported that patients treated with steroids alone or in combination with ASA have a higher incidence of coronary aneurysms, myocardial infarction, and death. In contrast, a more recent study of four IVIG-resistant KS patients suggested clinical improvement with high-dose pulse methylprednisolone (30 mg/kg once daily for 1–3 days). In 1999, Shinohara et al. reviewed prednisolone use in KS and concluded it was associated with shorter fever duration and lower aneurysm rates; however, this was a retrospective analysis involving patients who received outdated IVIG regimens not reflective of current U.S. practices.

Small case series have described the use of oral or intravenous corticosteroids in patients with IVIG-resistant disease (after at least two adequate IVIG doses), reporting symptomatic improvement without clear progression of coronary abnormalities. Due to the limited number of patients and lack of controlled data, the true impact of corticosteroids on coronary outcomes remains uncertain. Until well-designed controlled trials are available, corticosteroids should not be used routinely in the treatment of Kawasaki syndrome.